Lecture 3

1. **Regarding glucose:**
   a) All filtered glucose will appear in urine
   b) The tubular maximum for glucose (Tm is 475mg/ min)
   c) Glucose is completely reabsorbed in the PCT
   d) Glucose is reabsorbed through K^+- glucose co-transport

Answer: C

2. **Transport of glucose in the renal tubular cells occur via:**
   a) Primary Active transport
   b) Concentration gradient
   c) Facilitated diffusion
   d) Secondary active transport with sodium
   e) Secondary active transport with potassium

Answer: D

3. **The greatest fraction of filtered phosphate is reabsorbed in what portion of the nephron?**
   a) Proximal convoluted tubule
   b) Ascending limb of Loop of Henle
   c) Early part of Distal convoluted tubule
   d) P-cells of the collecting duct

Answer: A

4. **Regarding the transport of glucose by the renal tubules:**
   a) In a healthy person, the distal tubules reabsorb all of the filtered glucose
   b) Glucose is secreted into the urine in small quantities
   c) The transport maximum of glucose is about 36 mg/min
   d) Glucose transport by the renal tubules is linked to sodium transport

Answer: D

5. **Na^+ is reabsorbed from the basolateral surface of the renal epithelial cells by:**
   a) Na^+/H^+ exchange.
   b) Na-glucose cotransport
   c) Na^+-K^+ pump
   d) Facilitated diffusion

Answer: C

6. **Amino acids are almost completely reabsorbed from the glomerular filtrate via active transport in the:**
   a) Proximal convoluted tubule
   b) Loop of Henle.
   c) Distal convoluted tubule.
   d) Collecting duct

Answer: A

7. **Glucose is:**
   a) Filtered, reabsorbed, and secreted
   b) Filtered, reabsorbed, and not secreted
   c) Filtered, secreted, and not reabsorbed
   d) Filtered and neither reabsorbed or secreted

Answer: B

8. **Which of the following is most likely to be seen if a sample of fluid leaving the proximal tubule in a person with normal kidneys is analyzed?**
   a) It will be hypertonic compared with plasma
   b) It will have more K^+ than plasma
   c) It will have no amino acids
   d) It will have similar pH to that in ureter

Answer: C

9. **In the kidney, parathyroid hormone influences phosphate metabolism as it:**
   a) Favours the tubular reabsorption of filtered phosphate
   b) Inhibits the reabsorption of filtered phosphate
   c) Decreases the tubular load by significantly decreasing the glomerular filtration rate
   d) Increases the tubular load by decreasing calcium reabsorption.

Answer: B

10. **Phosphate reabsorption in the proximal convoluted tubules is:**
    a) Due to active co-transport with calcium ions
    b) Due to active co-transport with sodium ions
    c) Due to passive diffusion down its electrochemical gradient
    d) Inhibited by calcitonin

Answer: B

11. Which of the following is normally not found in the urine:

A) Water.
B) Waste products.
C) Sodium.
D) Albumin.

Answer: D

12. **Reabsorption of water through the tubules occurs by**
    a) osmosis.
    b) active transport.
    c) facilitated diffusion.
    d) all of these.

Answer: A

13. **What part of the nephron performs most of the reabsorption of materials from the filtrate?**
    a) The Bowman's capsule and glomerulus
    b) The loop of Henle (the nephron loop)
    c) The distal convoluted tubule and collecting duct
    d) The proximal convoluted tubule

Answer: D

14. **In which section of the renal tubule is most water reabsorbed**.
    a) Descending limb of the loop of Henle
    b) Ascending limb of the loop of Henle
    c) Collecting duct
    d) Proximal convoluted tubule

Answer: D

15. **The greatest fraction of filtered sodium is reabsorbed in what portion of the nephron?**
    a) Proximal tubule
    b) Loop of Henle
    c) Distal tubule
    d) Cortical collecting duct
    e) Medullary collecting duct

Answer: A

16. **The cells of the proximal convoluted tubules:**
    a) have very few mitochondria.

b) have many brush border.
c) responding mainly to ADH.
d) responding mainly to aldosterone.
e) are for reabsorption only.

Answer: B

17. **In the proximal convoluted tubules (PCT) all of the following occur EXCEPT:**
    a) active reabsorption of 65% of sodium.
    b) reabsorption of 65% of water under the control of ADH.
    c) complete active reabsorption of amino acids.
    d) complete active reabsorption of glucose.
    e) secondary active H^+ secretion.

Answer: B

18. **Sodium reabsorption in PCT is:**
    a) passive at the basolateral border.
    b) by secondary active transport at the basolateral border.
    c) by uniporter at the luminal border.
    d) about 90% of the filtered amount.
    e) about 65% of the filtered amount.

Answer: E

19. **Glucose is:**
    a) reabsorbed actively at the DCT.
    b) reabsorbed passively at the PCT.
    c) reabsorbed actively at the PCT.
    d) reabsorbed actively at the DCT & PCT.
    e) filtered but neither reabsorbed nor secreted by the nephron.

Answer: C

20. **Which of the following is reabsorbed passively in the PCT**:
    a) Na^+. b) K^+. c) glucose. d) urea. e) amino acids.

Answer: D